IMPERIAL COLLEGE LONDON

* Typo corrected on Page 3, Q. 2. by Course Leader @ 10:43.

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2019

EEE PART II: MEng, BEng and ACGI

Corrected Copy

ANALOGUE ELECTRONICS 2

Thursday, 13 June 10:00 am

Time allowed: 2:00 hours

There are THREE questions on this paper.

Answer ALL questions.
Q1 carries 40% of the marks. Questions 2 and 3 carry equal marks (30% each).

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : T. Constandinou
Second Marker(s) : C. Toumazou

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1. This question consists of 6 sub-questions. Each item can be answered in a few words or a short paragraph with use of a diagram and/or expression where helpful. Please answer all sub-questions and simplify expressions where possible stating any assumptions made.

a) Identify the key differences between MOSFET and BJT devices with respect to analogue circuit design. Describe when you would use one instead of the other using examples where appropriate. [5]

b) Draw the *small signal model* of an integrated BJT including parasitic capacitances and explain what limits its high frequency performance when used as a common emitter amplifier. [5]

c) Derive expressions (by inspection) for the voltage gains of the amplifier circuits shown below in Fig. 1.1. Include the effect of a finite output resistance. [10]

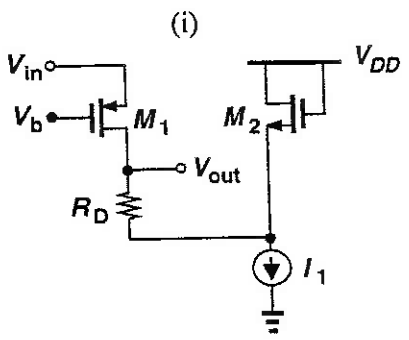


Fig. 1.1(a)

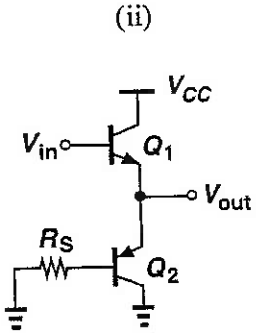


Fig. 1.1(b)

d) We wish to design the differentiator of Fig. 1.2 for a pole frequency of 434MHz. If the values of R_1 and C_1 cannot be lower than 15kΩ and 2.2nF, respectively, compute the required gain of the op amp. [5]

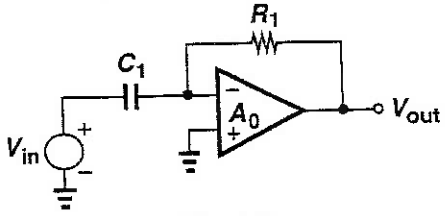


Fig. 1.2

e) Assuming perfect symmetry and $VA < \infty$, compute the differential voltage gain of the amplifier shown in Fig. 1.3. [5]

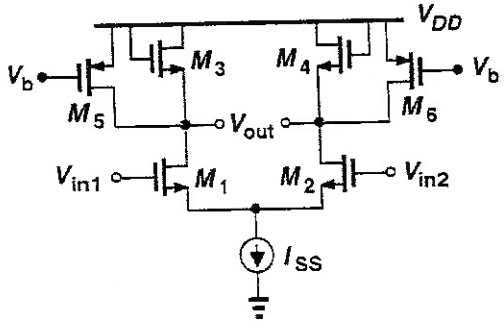


Fig. 1.3

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f) The amplifier shown below in Fig. 1.4 consists of a common-gate stage (M_1 and R_D) and a feedback network (R_1, R_2, and M_2). Assuming R_1 and R_2 is very large and λ=0, compute the closed-loop gain and I/O impedances.

[10]

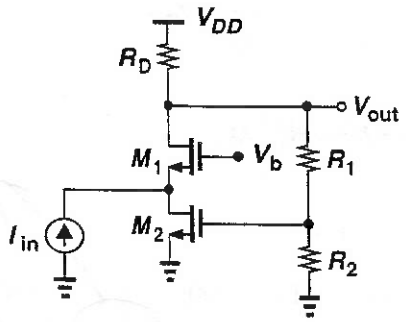


Fig. 1.4

2. The circuit shown below in Fig. 3.1 is a two-stage amplifier consisting of a common source amplifier (M_1) followed by a common gate (M_2) stage.

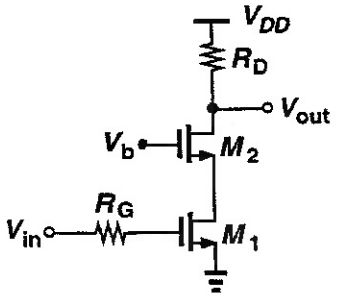


Fig. 3.1

Use the following parameters: $V_{DD} = 1.8V$, $(W/L)_1 = 100\mu m/0.18\mu m$, $(W/L)_2 = 50\mu m/0.18\mu m$, $I_{D1} = 2.5mA$, $\mu_n C_{ox} = 200\mu A/V^2$, $\lambda = 0$, $C_{ox} = 10fF/\mu m^2$, $C_{GS0}=C_{GS0}=C_0W$ where $C_0 = 0.2fF/\mu m$ denotes the gate-drain capacitance per unit width, $C_{DB}=C_{SB}=0$

a) Compare the frequency responses of a common source and common gate amplifier. [5]

b) Redraw the circuit above to: (i) identify any nodes that are associated with poles and (ii) include all the parasitic device capacitances (within M1-2). [5]

c) Redraw the circuit drawn in your answer to (b) to remove any redundant capacitances and lump together any parallel capacitances. Determine expressions for the different pole frequencies using Miller's theorem where appropriate. [5]

d) Determine an expression for the DC (low frequency) gain of the amplifier and evaluate for $R_D = 2k\Omega$. [5]

e) Evaluate the DC gain and pole frequencies, for $R_G = 10k\Omega$. [5]

f) Discuss how the high frequency performance can be improved. [5]

3. The circuit shown below in Fig. 3.1 is an integrated current reference circuit.

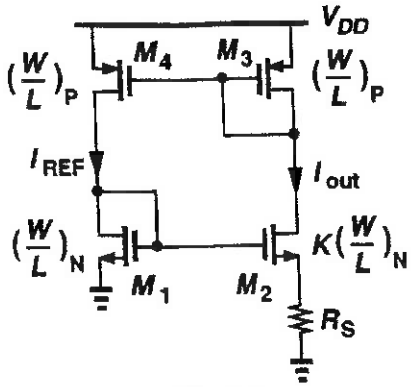


Fig. 3.1

Use the following parameters: $V_{DD} = 1.8V$, $\mu_n C_{ox} = 200\mu A/V^2$, $\mu_p C_{ox} = 100\mu A/V^2$, $\lambda_n=0.1$, $\lambda_p=0.2$

a) Discuss the challenges (i.e. non-idealities) of designing a current reference circuit – both for a circuit board implementation, and within an integrated circuit. [5]

b) Which of these challenges (in your answer to part (a)), if any, does the circuit shown in Fig. 3.1 address? How can the remaining challenges be addressed? [5]

c) Derive an expression for the output current, I_{out}, stating any assumptions made. Hint: this needs to be a function of $(W/L)_N$ and R_S. [5]

d) Design the circuit (i.e. determine the component values) such that an output current of $I_{out}=10\mu A$ is generated. [5]

e) It is required to extend the circuit to provide a secondary output current of $I_{out2}=250\mu A$.

(i) Describe how the circuit could be extended to generate the secondary output. [3]

(ii) Calculate the total power consumption of the revised circuit. [2]

f) Explain why the configuration shown in Fig. 3.2 may be preferable to that in Fig. 3.1. [5]

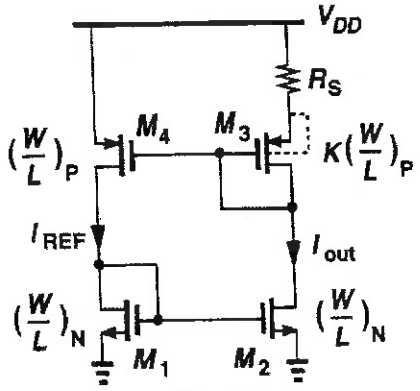


Fig. 3.2